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Management Tools 3D

version 2.1.2

requires: **uMMORPG Remastered** v2.36 and **TextMesh Pro**

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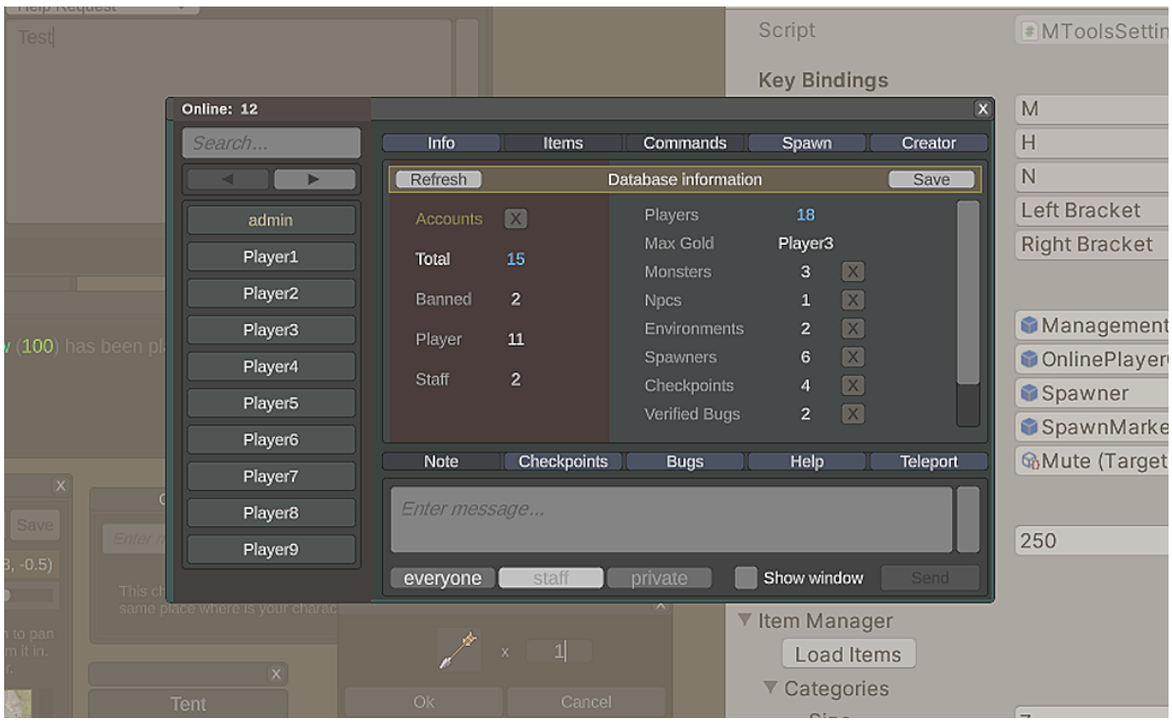
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MANAGEMENT TOOLS

Description

Management Tools has been designed as an addon for **uMMORPG Remastered**, which means that uMMORPG is required and must be installed first as well as **TextMesh Pro** (in Unity's top menu: Window > Package Manager). Management Tools is a collection of useful features and commands that allow you to manage your game, players and perform testing as a game master.



There are four levels of access that you can assign: 0 no access (banned), 1 player (standard player account), 2 special access (moderator), 3 owner (administrator).

Special access allows to see and do more, including creating spawn points or placing objects at runtime.

Go to the **table of contents** to find out information about all of the features.

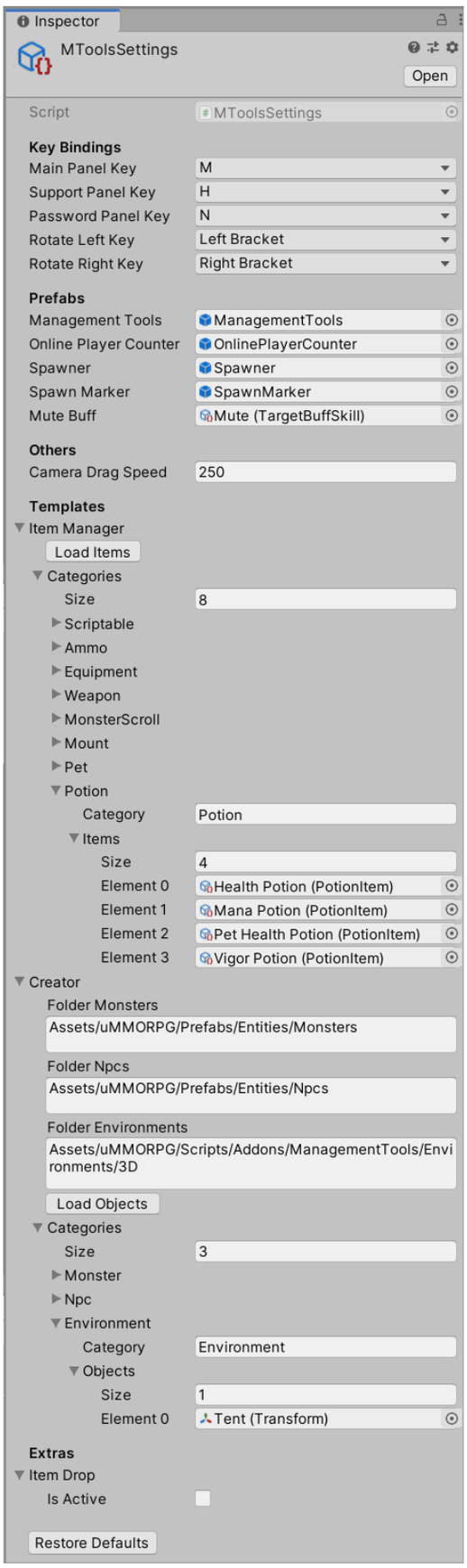
Preparing for installation

Make sure you have TextMesh Pro installed, if not you can find it in Unity's top menu: Window > Package Manager. Then click on Window > TextMeshPro > Import TMP Essential Resources.

Only then can you safely download this addon. After that use the **Setup Wizard** or install this addon manually as described in the **Installation Manual**.

MToolsSettings.asset stores all settings related to this addon. Configure it after the installation is complete.

The default settings can be restored by clicking on the Restore Defaults button.



How to use this panel

Use the `M` key to open the main panel (requires special access or owner permission). In the upper left corner, there is a counter showing how many players are currently in game. The player search field is located slightly below, then two arrows for turning pages and finally a list of all currently logged in players with 10 per page. On the bottom right there is a field for sending messages, and above there are two toolbars with buttons representing thematic panels. Some of them are active only if you have selected a player from the list. These buttons are: `Items`, `Commands`, `Note`, and `private`.

Use one of the 10 buttons on the top or bottom toolbar to open an additional panel by category. Go to the **table of contents** to find out information about all of them.

If the selected player logs out, some of the panels and the corresponding buttons will be deactivated.

If the access level will be lost the main panel will be deactivated.

LIST OF ALL CURRENTLY LOGGED IN PLAYERS



Description

On the left side, this is a list with 10 buttons per page that shows you all the users that are currently logged in. Click on the button to select a player which will unlock some additional options. If the selected player logs out, some of the panels and the corresponding buttons will be deactivated.

Keyword searching

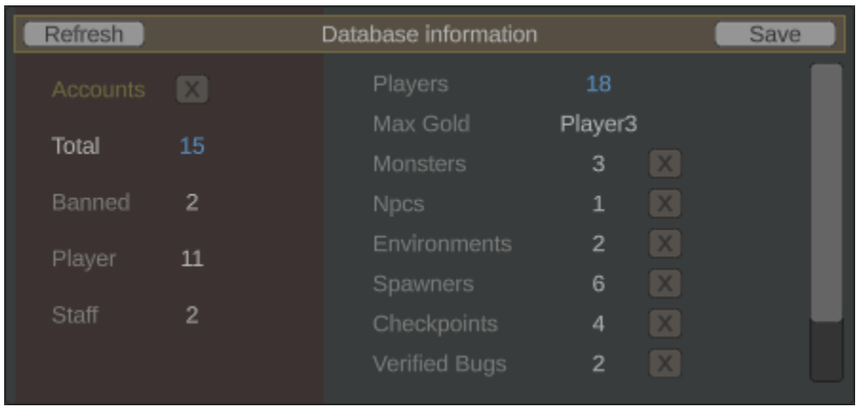
Type a single letter or word, and the list and other things will be filtered using your search term. Search results are also updated when a player logs in or out.

DATABASE INFORMATION

Description

Allows you to see the number of players, accounts and other things saved in the database. Each time you open this panel or click on the `Refresh` button, all data will be updated from the database.

Use the `Info` button to open this panel.



You can save the current world state with the `Save` button without waiting for the regular save cycle.

Ability to delete data from the database and remove objects from the game.

> **Additional field with the number of items and the delete option. To activate it go to MToolsSeetings.asset and check "Extras/item Drop Addon/isActive" in the Inspector window.**
>
> **Requires Item Drop for the uMMORPG.**

Deleting data from database

Use one of the delete `x` buttons next to a table name to clear the table contents, and the corresponding objects from the game. Confirm your decision by clicking the `ok` button.

Owners can also delete all accounts with characters and other related tables except other owner accounts. This option `x` is next to the accounts header, but is enabled only for the owner (access level 3).

ITEM MANAGER

Description

All items from the game divided by category. To see them use the `Items` button, which is only active, if a player has been selected. Now you can give him any item.

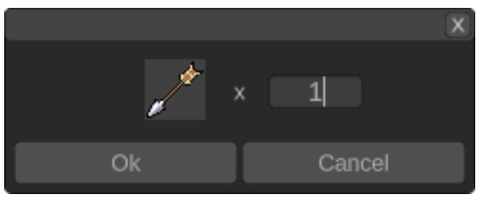


If the selected player logs out, this panel and button will be deactivated.

Scrollbars will appear automatically for the categories and items.

> **Item name color based on the rarity of the item.**
>
> **Requires Item Drop for the uMMORPG.**

How to give an item



Click the `+` button (next to item name). If the item is stackable, an additional window will open where you can choose how many items you want to send to the player's inventory. The player will only receive the item(s) if there is enough space in their inventory.

> **You can send up to one "stack" of an item at a time. The size of a stack is dependent on the maxStack value of the ScriptableItem. When maxStack is greater than 1, you will be prompted to set a quantity from 1 to maxStack to send.**

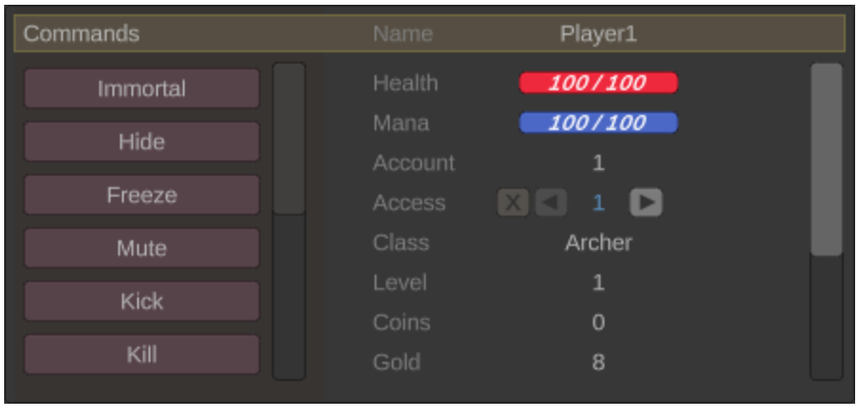
Keyword searching

Type a single letter or word, and the list of items will be updated. You do not need to choose a category for this to work.

COMMANDS

Description

Displays the current information about the selected player, as well as commands that you can use. The displayed data is updated in a small time interval.



Click the `Commands` button, active only if a player has been selected.

If the selected player logs out, this panel and button will be deactivated.

Access

Use the `arrows` placed in the access field to change the access level of the selected player between 1-2 and confirm your decision in the open window. You cannot set access level 3 from the management panel.

Use the `x` button to ban them. They will be removed from the game immediately (requires access level 3). Not active, if you select your character.

Four levels of access:	
0	banned
1	player
2	special access
3	owner

Immortal

Selected player will be immortal. Other players and enemies cannot deal any damage to him. Now the button will show `Mortal`, use again to undo this effect.

Hide

Selected player will be invisible for everyone (including monsters). Now the button will show `Unhide`, use again to make them visible again.

Freeze

Selected player will be unable to move. Movement is immediately interrupted and the target canceled. Now the button will show `Unfreeze`, use again to unfreeze them.

Mute

Selected player will be muted for 15 minutes. Typing and sending chat messages is disabled. Now the button will show `Unmute`, use again to undo this effect.

> **You can change this time by editing Mute.asset in Assets/uMMORPG/Scripts/Addons/ManagementTools/Resources**

Kick

Selected player will be kicked from the server. Only players (access level 1) can be kicked. Not active, if you select your character.

Kill

Selected player will be killed. Now the button will show `Resurrect`, use again to resurrect them. They will resurrect where they died with a full health and mana bar.

Fetch

Selected player will be moved to your position. Not active, if you select your character. No effect if you are in a dungeon instance.

Go To

Your character will be moved to the selected player's position. His movement is immediately interrupted. Not active, if you select your character. No effect if you are in a dungeon instance.

Gold

Type a number between 1-1000 into the input field, and click to give that amount of gold to the selected player. Only working when you have something in the input field.

Coins

Type a number between 1-1000 into the input field, and click to give that amount of coins to the selected player. Only working when you have something in the input field.

Exp

Type a number between 1-1000 into the input field, and click to give that amount of experience to the selected player. Only working when you have something in the input field.

Level

Type a number between 1-100 into the input field, and click to set the player's level. However, the level will be increased only to the maximum as in the player's prefab. The health and mana bar will fill up immediately. Only working when you have something in the input field.

Strength

Type a number between 0-100 into the input field, and click to set strength of the selected player. The health bar will fill up immediately. Only working when you have something in the input field.

Intelligence

Type a number between 0-100 into the input field, and click to set intelligence of the selected player. The mana bar will fill up immediately. Only working when you have something in the input field.

CREATOR

Description

All spawnable objects from the game divided by category: monsters, npcs and environments. To see them use the `Creator` button. Now you can choose categories and objects.



Scrollbars will appear automatically for the categories and objects.

This allows you to spawn new objects at runtime.

Sample objects

Once this addon is installed and configured go to Assets/uMMORPG/Scripts/Addons/ManagementTools/Environments/3D to see a sample environment object. Look at how it has been prepared.

> **Tent.prefab: has 3 invisible walls (Empty GameObjects with NavMeshObstacle as a box shape) blocking the transition where the size and position has been set manually.**

All new environment prefabs you want to spawn at runtime must be placed in a folder as in the path assigned in the MToolsSettings.asset, to easy-add them to the array. They also need to be prepared like the sample ones before you will use the `Load Objects` button again.

There is also another way to create environment objects, described in the next header.

Standard and special version

Choose the standard version (Assets/uMMORPG/Scripts/Addons/ManagementTools/Environments/3D) if you do not need more shapes than a box or capsule to block the movement in the game.

> **Tent2.prefab: no invisible walls but it has a NavMeshSourceTag component and requires scene editing as described in the *ADDITIONAL STEPS, ONLY FOR THE SPECIAL VERSION.**

In the special version (Assets/uMMORPG/Scripts/Addons/ManagementTools/Environments/Special) NavMesh is baked in a new way using NavMeshComponents created by Unity Technologies (not supplied with the standard Unity Editor installer for now). No limit to the shape and it is generated automatically.

How to prepare the default scene for a special version is described at the end of the Installation Manual ***ADDITIONAL STEPS, ONLY FOR THE SPECIAL VERSION**.

How to create an object

Click the `+` button (next to object name), additional field with the selected object name will be open on the main panel header. If this field is there, the selected object is visible but not spawned yet and follows the cursor position. You can use `[` and `]` to rotate it before spawn. Use the middle click `( Scroll Wheel )` to place this object on the game at the cursor position. NavMesh is the only clickable area.

All of these objects will receive their unique ID and will be saved to the database and loaded after server restart.

Use delete button `x` (next to the name of object on the top) to cancel placing object using `Scroll Wheel`.

> **Do not build your entire map using environments from this panel, instead of the Unity Editor. Use them for special events.**

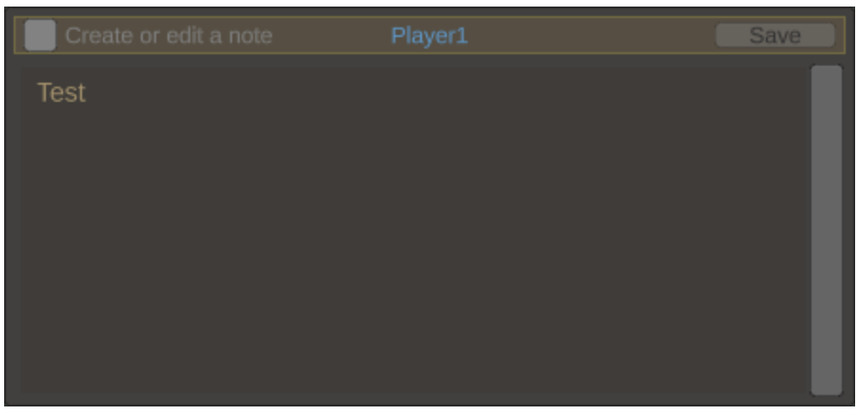
Keyword searching

Type a single letter or word, and the list of objects will be updated. You do not need to choose a category for this to work.

NOTE

Description

You can write and edit a note about the selected player. Select him on the list of players and use the `Note` button. Each time you open this panel, a player's note will be created or loaded from the database. If the note is empty you will see an empty field. If any of the staff members saves a note and this panel is open with the same player selected, others will see these changes too.



If the selected player logs out, this panel and button will be deactivated.

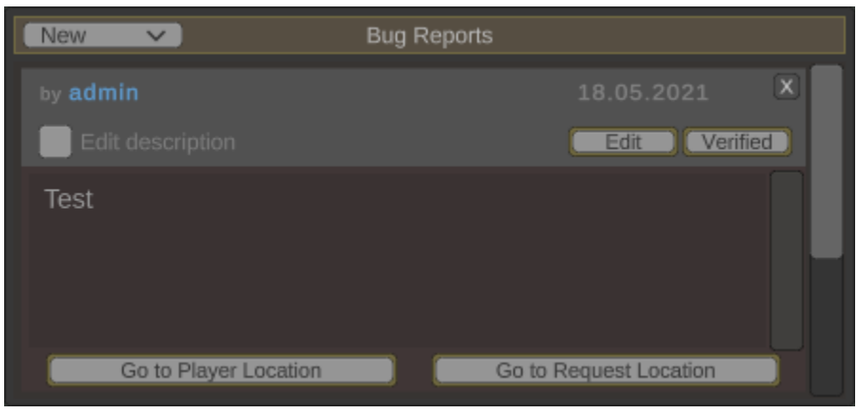
Editing a note

Use the switch in the left corner to enter the edit mode. Write whenever you want and confirm with the `Save` button.

BUG REPORTS

Description

This feature allows players to file a bug report when they encounter a bug. Encountered bugs can be reported in game by everyone. New bugs are assigned to the “new” category. To see them, use the `Bugs` button. Other categories are: verified and fixed. Each time you open this panel or change a bug category by clicking on the dropdown in the upper left corner, all bugs will be loaded from the database by that category as a list of buttons with the author's name and date of sending.



If this panel is open when someone reports a new bug, it will be immediately visible in the list. This also applies to other activities, such as changing status or deleting a bug report.

Click any of the buttons and the mini panel with the bug report will be open. You will also see two buttons: `Go to Player Location` and `Go to Request Location` to assist in responding or recreating the bug.

Editing an existing bug report

New reports should be verified first. If you are sure of your decision, click the `Verified` button. This report will be moved to the verified category. Otherwise decline it by clicking the delete button `x`. Proceed similarly if the issue has been resolved by opening the verified category, click any of the reports and use the `Fixed` button. Now the report will be placed in the fixed category.

The content of the report can be edited. To do this use the toggle in the left corner to enter edit mode. After making changes, use the `Edit` button to save your changes.

How to report a bug

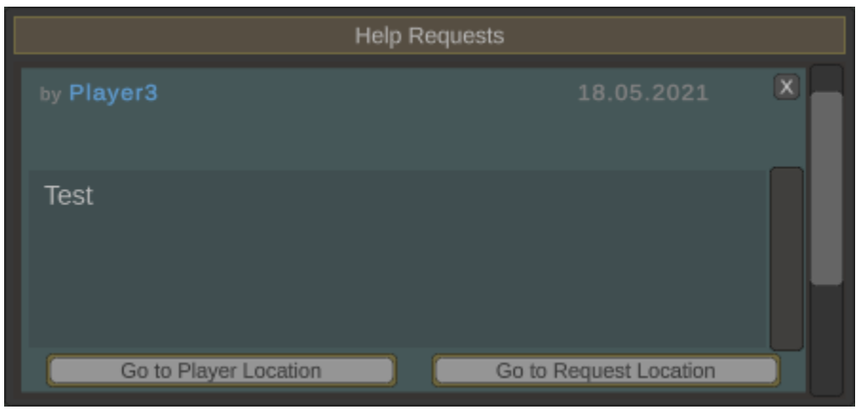
Any player can report a bug by pressing the `H` key to open the support panel. Change the category to bug report by clicking on the dropdown in the upper left corner. Now describe what you have noticed and confirm with the `Send` button. You will be informed in the chat about sending a bug report. Staff members will also get notification.

> **Players are expected to report all bugs, describing every step needed to reproduce the issue which should help shorten the time needed to recreate the issue and fix it.**

HELP REQUESTS

Description

Players can send requests for help. To see the help requests, use the `Help` button. Each time you open this panel all help requests will be loaded from the database as a list of buttons with the author's name and the date/time it was sent.

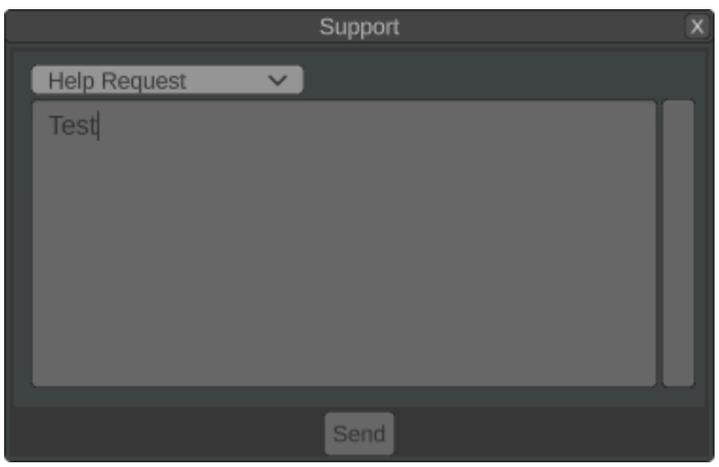


If this panel is open when someone requests help, it will be immediately visible in the list. This also applies to deleting a help request.

Click any of the buttons and the mini panel with the message will be open. You will also see two buttons: `Go to Player Location` and `Go to Request Location` to allow for faster resolution and support. If you want to get rid of the request, use the delete button `x`.

How to send a request for help

Any player can submit a help request by pressing the `H` key to open the support panel. Choose the help category by clicking on the dropdown in the upper left corner. Now describe your problem and confirm with the `Send` button. You will be informed in the chat about sending your help request and your number. Staff members will also get notification.



TELEPORT

Description

Use the `Teleport` button to open a minimap. Hold down the right mouse button to pan the map. Use the scrollbar to zoom it in. Each time you click on the minimap your character will teleport to that position. NavMesh is the only clickable area. Use the `Show My Location` button to show your position on the minimap.



You can show or hide the default minimap markers using the toggle at the top.

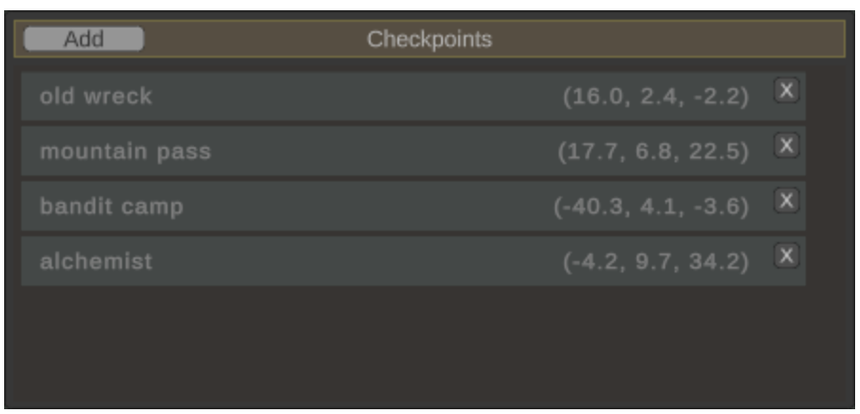
> **This feature can be combined with the checkpoints section to save time teleporting to predefined locations.**

CHECKPOINTS

Description

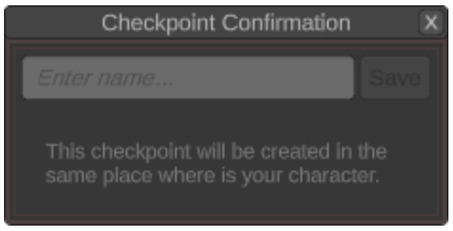
Checkpoints can be created by characters with special access (level 2).

To see them, use the `Checkpoints` button. Each time you open this panel all checkpoints will be loaded from the database as a list of buttons with assigned name and position. If you want to remove any of the checkpoints, use the delete button `x`.



If any of the staff members creates a new checkpoint or deletes an existing checkpoint, other staff members will see the change right away.

How to create a new checkpoint

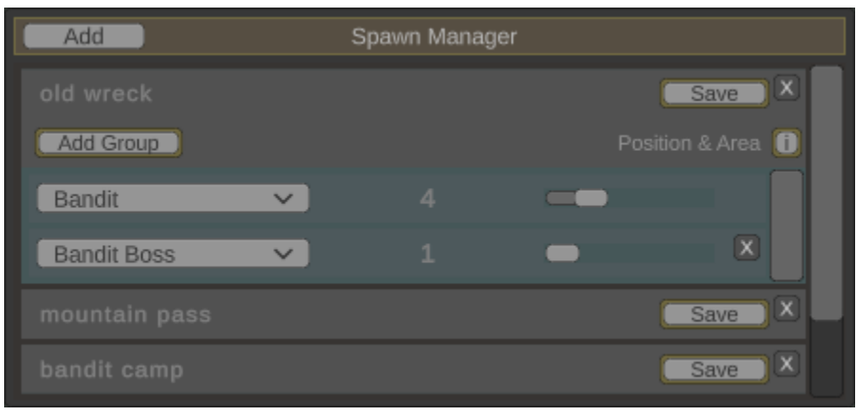


Use the `Add` button of the checkpoints panel. You will see a simple window to enter the checkpoint's name, the position will be the same as your character's current position. Confirm your choice with the `Save` button.

SPAWN MANAGER

Description

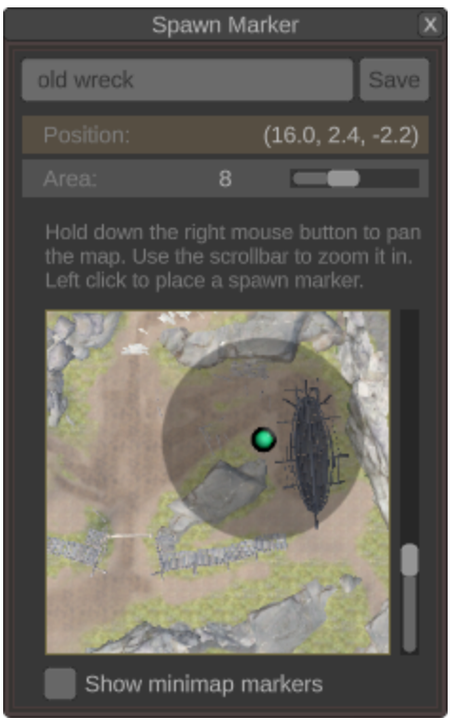
Creating a spawner is a very useful feature, even more so that it does not require using the Unity Editor. The entire panel is easy to use and for sure will save you a lot of time. One spawn point can generate more than one selected group of monsters.



Use the `Spawn` button to open it. Each time you open this panel all spawn points data will be loaded from the database as a list of buttons.

If any of the staff members activates a new spawner or make a change to the existing spawners, other staff members will see these changes right away.

How to create a new spawner



Click the `Add` button and another panel will be opened.

There you can choose a name, position and area. Hold down the right mouse button to pan the map. Use the scrollbar to zoom it in. Left click to place a spawn marker. It can be placed only on the NavMesh. Use the `Position` button to show spawn marker position on the minimap. Choose the size of the area. It will be visible on the marker. You can show or hide the default minimap markers using the switch at the bottom. Confirm your choices with the `Save` button.

Click on the newly created button in the list. First monster group is always active and cannot be deleted. Choose enemy type and the number to spawn. You can create more groups, by clicking the `Add Group` button. Each spawn button and its spawn groups have their own delete button `x`. At the end use the

`Save` button and every spawn point will be saved or updated. Congratulations, the new spawner is active!

> **Change the Respawn Time in the Inspector of your monster prefabs. They should not respawn too quickly.**

Editing existing spawner

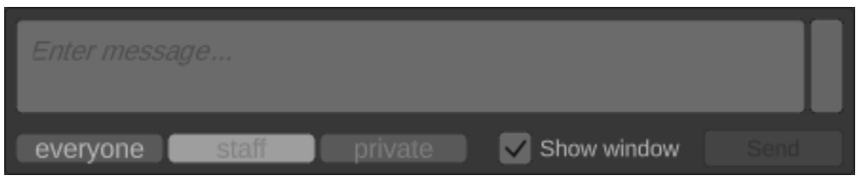
Use the button labeled as `i` (next to Position & Area) and do the same as using the `Add` button.

Always remember to use the `Save` button at the end (on the list). If a spawn point is deleted or edited, the currently spawned monsters will be removed from the game.

INFO MESSENGER

Description

In the lower right corner of the main panel there is a small panel for sending a message.

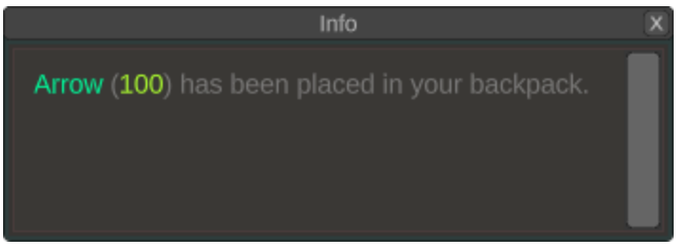


You can send a mass message to all logged in players, staff or to the currently selected player.

No restrictions in the player's location. They can be in your visible range or not, it does not matter.

Info window

You can enable the `Show Window` option and all messages will be shown not only on the chat but there will also be a small popup dialog. This applies to both sending messages and other actions such as commands.



How to send a message

This panel is always visible. To choose where the message will be sent, turn on one of the three buttons: `Everyone` (selected by default), `Staff` (always active, players without special access will not see these messages), `Private` is only active if a player is selected.

Enter a message and use the `Send` button, which can be used only if the message input field is not empty. After sending a message, entered text will be cleared.

If the selected player logs out the `Private` button will be disabled and the `Everyone` button re-enabled as the default.

> **You should always inform your players in game about upcoming server restart to give them time to finish certain actions and logout in a safe place.**

MANIPULATION TOOLS

Description



Allows you to rotate monsters, npcs or environments created with the creator section and also kill or remove them from the game (requires access level 2 or 3). Use the `Save` button at the end of editing (changing the rotation and height) and all changes will be saved to the database and loaded after server restart.

Editing existing object

Right click on the monster, npc or environment to open this panel (requires unique ID for all functions). However, you can click on monsters without an unique ID. In this situation there will be no `Delete` or `Save` button, but you can rotate monsters without saving to the database or kill them with the `Kill` button.

Use the `arrows` to change the rotation or position y of the selected object. Click the `Save` button at the end to save all changes to the database.

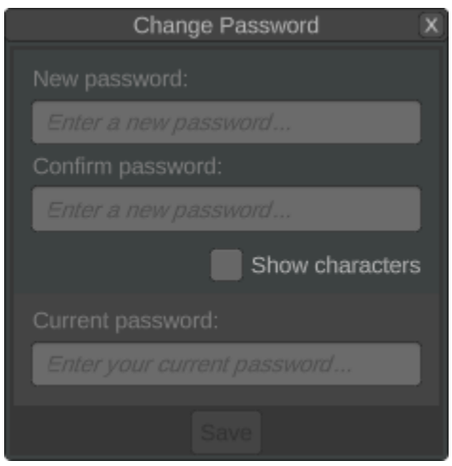
Use the `Kill` button to kill the selected monster.

Use the `Delete` button to remove the selected object from the game and delete it from the database.

PASSWORD CHANGE

Description

Each player can change his password in game.



How to change your password

Use the `N` Key to display the password change window. Enter a new password, confirm it and also enter the current one for security. Confirm your choice with the `Save` button, active only if all text fields have been filled correctly.

> **Switch on “Show Characters” if you wanna see all characters in the input fields, except the current password.**

SETUP WIZARD

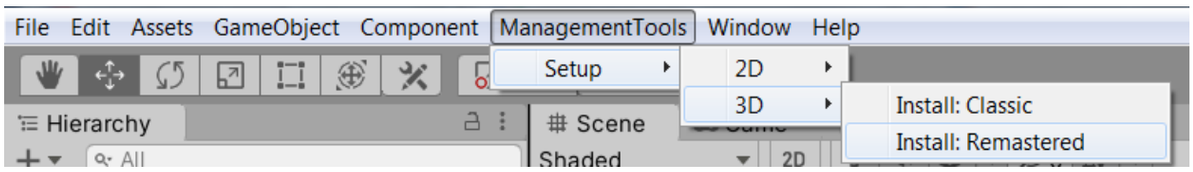
Description

Version 2.1.0 comes with the Setup Wizard. Now you can install everything in a quick and easy way.

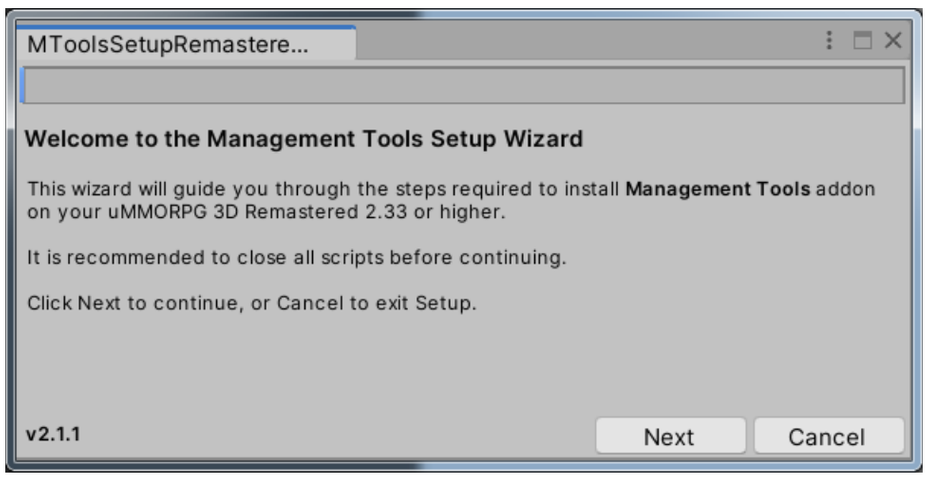
A few clicks to install the entire addon or uninstall as needed. Just click a button to replace or add code, components and prefabs. However, if you have previously made any changes that can conflict with the installer you must install this addon manually. See the **INSTALLATION MANUAL**.

How to install

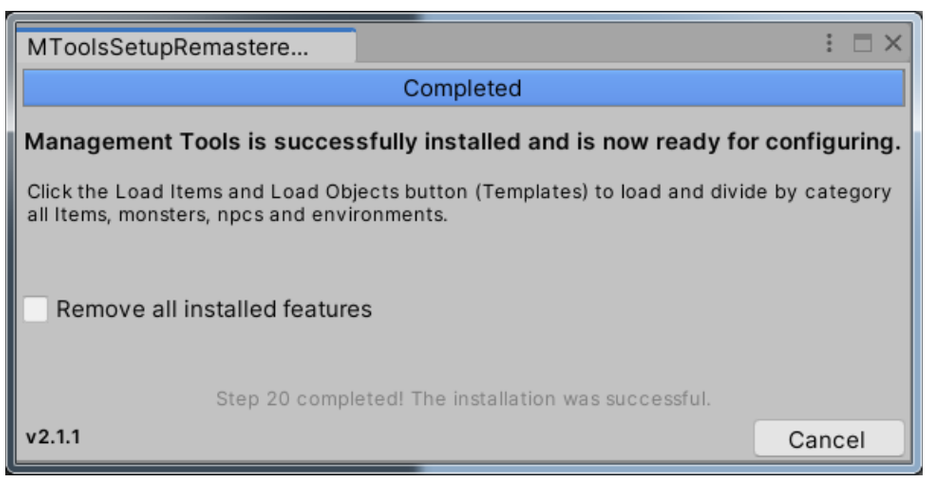
In Unity's top menu: ManagementTools > Setup > 3D click and choose the remastered version of uMMORPG (Install: Remastered).



This wizard will guide you through the steps required to install Management Tools addon on your uMMORPG 3D.



Follow the instructions displayed on the window and click `Finish` at the end. If everything works, you can skip the **INSTALLATION MANUAL**.

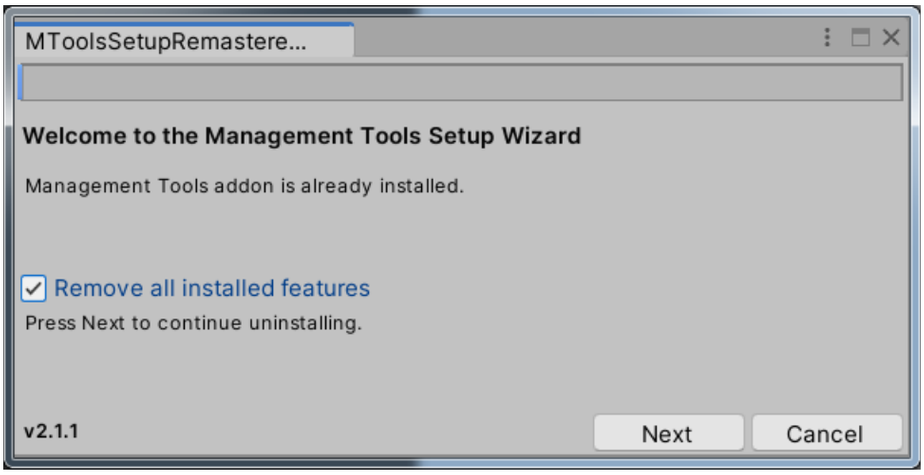


After the installation is completed, the MToolsSettings.asset Inspector window will open automatically, where you can configure all the settings.

How to uninstall

Open the Setup Wizard in the same way as for installation. In Unity's top menu: ManagementTools > Setup > 3D click and choose the remastered version of uMMORPG (Install: Remastered).

If this addon is already installed you can check "Remove all installed features". Follow the instructions displayed on the window and click `Finish` at the end.



After uninstalling, you can safely remove the ManagementTools folder from your project: Assets/uMMORPG/Scripts/Addons/ManagementTools.

INSTALLATION MANUAL (uMMORPG 3D Remastered)

Step 1

Edit the UICharacterCreation.cs Script in “Assets/uMMORPG/Scripts/_UI”

Comment this line

```
            //gameMasterToggle.gameObject.SetActive(NetworkServer.localClientActive);
```

Comment this line

```
                    //gameMaster = gameMasterToggle.isOn
```

Step 2

Edit the UIChat.cs Script in “Assets/uMMORPG/Scripts/_UI”

> **void Update()**

Find this line

```
            if (Utils.AnyKeyDown(activationKeys) && !eatActivation)
```

and change to

```
            if (Utils.AnyKeyDown(activationKeys) && !eatActivation && !UIManagement.singleton.canvas.enabled)
```

Step 3

Edit the UIGameMasterTool.cs Script in “Assets/uMMORPG/Scripts/_UI”

> **void Update()**

Find this line

```
            if (panel.activeSelf) && player.isGameMaster)
```

and change to

```
            if (panel.activeSelf)
```

Step 4

Edit the UIKeepInScreen.cs Script in “Assets/uMMORPG/Scripts/_UI”

Add

```
    void Awake()
    {
        if (!rectTransform)
        {
            rectTransform = GetComponent<RectTransform>();
        }
    }
```

Step 5

Edit the UIShortcuts.cs Script in “Assets/uMMORPG/Scripts/_UI”

Comment these lines

```
            /*
            gameMasterButton.gameObject.SetActive(player.isGameMaster);
            gameMasterButton.onClick.SetListener(() => {
                gameMasterPanel.SetActive(!gameMasterPanel.activeSelf);
            });
            */
```

Step 6

Edit the UIUtils.cs Script in “Assets/uMMORPG/Scripts/_UI”

Add this line on the top of the script

```
using TMPro;
```

public static bool AnyInputActive()

Find this line

```
            if (sel is InputField inputField && inputField.isFocused)
```

and change to

```
            if ((sel is InputField && ((InputField)sel).isFocused) || (sel is TMP_InputField &&
((TMP_InputField)sel).isFocused))
```

Step 7

Edit the Database.cs Script in “Assets/uMMORPG/Scripts”

class accounts

Comment this line

```
        //public bool banned { get; set; }
```

and insert below

```
        public int AccessLevel { get; set; }
```

class characters

Comment this line

```
        //public bool gamemaster { get; set; }
```

public bool TryLogin(string account, string password)

Comment this line

```
                //connection.Insert(new accounts{ name=account, password=password, created=DateTime.UtcNow,
lastLogin=DateTime.Now, banned=false});
```

and insert below

```
            if (account == "admin")
                connection.Insert(new accounts
                {
                    name = account,
                    password = password,
                    created = DateTime.UtcNow,
                    lastlogin = DateTime.Now,
                    AccessLevel = 3
                });
            else
                connection.Insert(new accounts
                {
                    name = account,
                    password = password,
                    created = DateTime.UtcNow,
                    lastlogin = DateTime.Now,
                    AccessLevel = 1
                });
```

Find this line

```
            if (connection.FindWithQuery<accounts>("SELECT * FROM accounts WHERE name=? AND password=? and banned=0",
account, password) != null)
```

and change “banned” to “NOT AccessLevel”

```
            if (connection.FindWithQuery<accounts>("SELECT * FROM accounts WHERE name=? AND password=? and NOT
AccessLevel=0", account, password) != null)
```

public GameObject CharacterLoad(string characterName, List<Player> prefabs, bool isPreview)

Comment this line

```
            //player.isGameMaster                       = row.gamemaster;
```

public CharacterSave(Player player, bool online, bool useTranscation = true)

Comment this line

```
            //gamemaster = player.isGameMaster,
```

Step 8

Edit the Entity.cs Script in “Assets/uMMORPG/Scripts”

public bool IsHidden() =>

Find this line

```
netIdentity.visible == Visibility.ForceHidden;
```

and change to

```
netIdentity.visible == Visibility.ForceHidden || invisible;
```

Step 9

Edit the Instance.cs Script in “Assets/uMMORPG/Scripts”

> **void Update()**

Add these lines below player.movement.Warp(spawn.position);

```
#if UMMORPG_3D_R
                            player.DungeonControl = false;
#endif
```

Step 10

Edit the NetworkManagerMMO.cs Script in “Assets/uMMORPG/Scripts”

> **void LoadPreview(GameObject prefab, Transform location, int selectionIndex, CharacterAvailableMsg.CharacterPreview character)**

Comment this line

```
        //player.isGameMaster = character.isGameMaster;
```

> **Player CreateCharacter(GameObject classPrefab, string characterName, string account, bool gameMaster)**

Remove bool gameMaster from this line

```
    Player CreateCharacter(GameObject classPrefab, string characterName, string account)
```

Comment this line

```
        //player.isGameMaster = gameMaster;
```

> **void OnServerCharacterCreate(NetworkConnection conn, CharacterCreateMsg message)**

Comment these lines

```
                            /*
                            if (message.gameMaster == false ||
                                conn == NetworkServer.localConnection)
                            {
                                // create new character based on the prefab.
                                Player player = CreateCharacter(playerClasses[message.classIndex].gameObject,
message.name, account, message.gameMaster);

                                // addon system hooks
                                onServerCharacterCreate.Invoke(message, player);

                                // save the player
                                Database.singleton.CharacterSave(player, false);
                                Destroy(player.gameObject);

                                // send available characters list again, causing
                                // the client to switch to the character
                                // selection scene again
                                conn.Send(MakeCharactersAvailableMessage(account));
                            }
                            else
                            {
                                //Debug.Log("character insufficient permissions for GM request: " + conn);  <- don't
show on live server
```

```
                            ServerSendError(conn, "insufficient permissions", false);
                        }
                        */
```

and insert below

```
                        // create new character based on the prefab.
                        Player player = CreateCharacter(playerClasses[message.classIndex].gameObject,
message.name, account);

                        // addon system hooks
                        onServerCharacterCreate.Invoke(message, player);

                        // save the player
                        Database.singleton.CharacterSave(player, false);
                        Destroy(player.gameObject);

                        // send available characters list again, causing
                        // the client to switch to the character
                        // selection scene again
                        conn.Send(MakeCharactersAvailableMessage(account));
```

IEnumerator<WaitForSeconds> DoServerDisconnect(NetworkConnection conn, float delay)

Comment these lines

```
            //Database.singleton.CharacterSave(conn.identity.GetComponent<Player>(), false);
            //Debug.Log("saved:" + conn.identity.name);
```

and insert below

```
            if (conn.identity.TryGetComponent(out Player player))
            {
                if (player.canBeSaved)
                {
                    Database.singleton.CharacterSave(player, false);
                    //Debug.Log("saved:" + conn.identity.name);
                }
            }
```

Step 11

Edit the NetworkMessages.cs Script in “Assets/uMMORPG/Scripts”

public partial struct CharacterCreateMsg : NetworkMessage

Comment this line

```
    //public bool gameMaster; // only allowed if host connection!
```

public partial struct CharacterPreview

Comment this line

```
        //public bool isGameMaster; // for nameoverlay prefix in preview!
```

public void Load(List<Player> players)

Comment this line

```
                //isGameMaster = player.isGameMaster,
```

Step 12

Edit the PlayerGameMasterTool.cs Script in “Assets/uMMORPG/Scripts”

> **public override void OnStartServer()**

Comment this line

```
        //if (!player.isGameMaster) return;
```

Step 13

Edit the Player.cs Script in “Assets/uMMORPG/Scripts”

Comment this line

```
    //public string nameOverlayGameMasterPrefix = "[GM] ";
```

Comment this line

```
    //[SyncVar] public bool isGameMaster;
```

> **public override void OnStartLocalPlayer()**

Insert this line at the end

```
        OnStartLocalPlayer_Mute();
```

> **protected override void Start()**

Insert this line at the end

```
        Start_StaffMembersUpdate();
```

> **void OnDestroy()**

Insert this line at the end

```
        OnDestroy_StaffMembersUpdate();
```

> **string UpdateServer_DEAD()**

Add this at the beginning

```
        if (EventResurrect())
        {
            Resurrect();
            return "IDLE";
        }
```

public bool IsMovementAllowed()

Find this line

```
        return (state == "IDLE" || state == "MOVING" || castingAndAllowed) &&
               !isLocalPlayerTyping;
```

and change to

```
        return (state == "IDLE" || state == "MOVING" || castingAndAllowed) && !isLocalPlayerTyping && !IsFrozen();
```

protected override void UpdateClient()

Add this line at the beginning

```
        UpdateClient_SetObjectViaPosition();
```

Step 14

Edit the PlayerNavMeshMovement.cs Script in “Assets/uMMORPG/Scripts/MovementSystems”

void MoveClick()

Find this line

```
                if (!hit.transform.GetComponent<Entity>())
```

and change to

```
                if (!hit.transform.GetComponent<Entity>() && !hit.transform.CompareTag("Environment"))
```

Step 15

Edit the Portal.cs Script in “Assets/uMMORPG/Scripts”

void OnPortal(Player player)

Find this line

```
        if (destination != null)
            player.movement.Warp(destination.position);
```

and change to

```
        if (destination != null)
        {
            player.movement.Warp(destination.position);
#if UMMORPG_3D_R
            player.DungeonControl = false;
#endif
        }
```

Step 16

Edit the PortalToInstance.cs Script in “Assets/uMMORPG/Scripts”

> **void OnPortal(Player player)**

Find this line

```
                if (player.isServer) player.movement.Warp(existingInstance.entry.position);
```

and change to

```
                if (player.isServer)
                {
                    player.movement.Warp(existingInstance.entry.position);
#if UMMORPG_3D_R
                    player.DungeonControl = true;
#endif
                }
```

Find this line

```
                    if (player.isServer) player.movement.Warp(instance.entry.position);
```

and change to

```
                    if (player.isServer)
                    {
                        player.movement.Warp(instance.entry.position);
#if UMMORPG_3D_R
                        player.DungeonControl = true;
#endif
                    }
```

Step 17

Create a new Tag. Click on the Inspector window Tag list, then Add Tag Environment.

Create a new Layer. Click on the Inspector window Layer list, then Add Layer Environment.

Step 18

Edit all player prefabs in “Assets/uMMORPG/Prefabs/Entities/Players”

Turn off the PlayerGameMasterTool.cs component.

Set the CollisionLayerMask as Default and Environment in the CharacterController2k.cs component.

Step 19

Edit the Inspector window “Assets/uMMORPG/Scripts/Addons/ManagementTools/Panels/SpawnManager/SpawnMarker.prefab”

Create a new Layer. Click on the Inspector window Layer list, then Add Layer Pin and set it on this prefab (select “Yes, change children”).

Step 20

Edit the Main Camera's Inspector window/Camera/Culling Mask in the World scene.

Edit the MiniMap Camera's Inspector window/Camera/Culling Mask in the World scene.

Edit the Inspector window/Camera/Culling Mask "Assets/uMMORPG/Scripts/Addons/ManagementTools/Prefabs/ManagementTools.prefab/Teleport Camera"

Uncheck Pin from the Culling Mask list, then make sure it is selected in your Spawn Camera (above Teleport Camera).

Make sure the Environment is selected in the Culling Mask list in your Main Camera, MiniMap Camera, Spawn Camera, Teleport Camera.

Step 21

Drag and drop ManagementTools.prefab from "Assets/uMMORPG/Scripts/Addons/ManagementTools/Prefabs" to the World scene.

Drag and drop OnlinePlayerCounter.prefab from "Assets/uMMORPG/Scripts/Addons/ManagementTools/OnlinePlayerCounter" to the World scene **and** to the NetworkManager Registered Spawnable Prefabs.

Drag and drop Spawner.prefab from "Assets/uMMORPG/Scripts/Addons/ManagementTools/Resources" to the NetworkManager Registered Spawnable Prefabs.

Step 22

Edit the GameMasterTool in "World/Canvas/GameMasterTool" on the World scene.

Turn off the UIGameMasterTool.cs component.

Step 23

Edit the CharacterCreation/CharacterCreationPanel/CenterPanel/ToggleGameMaster in "World/Canvas/CharacterCreation" on the World scene.

Turn off the ToggleGameMaster GameObject, navigate to the Inspector and uncheck the checkbox next to the GameObject's name.

Step 24

Edit the Project Settings from Unity's main menu "Edit/Project Settings/Player/Configuration/Scripting Define Symbols"

Insert ;UMMORPG_3D_R at the end and push the Enter button.

Now all lines of code related with the 3D remastered version are active.

Step 25

Go to the ***ADDITIONAL STEPS, ONLY FOR THE SPECIAL VERSION** if you want to use it, if not do as described below and in the next step.

Configure the MToolsSettings.asset in "Assets/uMMORPG/Scripts/Addons/ManagementTools/Resources"

Click the Load Items button (Templates/Item Manager). All items will be loaded automatically and divided by category. Your project will also be saved.

Make sure the Environments Folder path ends with "3D" (Templates/Creator). Click the Load Objects button. All monsters, npcs and environments will be loaded automatically and divided by category, then assigned to NetworkManager Registered Spawnable Prefabs. The Environment Tag and Layer will be set automatically on all prefabs that are in the destination folder. Your project will also be saved.

You can always click the Restore Defaults button (at the bottom in the Inspector window) to restore default settings. Your project will also be saved.

> **MToolsSettings.asset stores all settings related to this addon. You can configure it however you want.**
>
> **You will also find an additional option here:**
> **Item Drop turn on if you have Item Drop. Now an additional field with items in the Database Information is active.**

Step 26

Create new account "admin", will be with the owner access. Now every character on this account has the owner access.

Use the M key to open the Management Tools. Now you can change all logged players access in the panel to 0, 1 or 2. Go to the **table of contents** to find out information about all other features.

> **How to set the access level outside the game:**
> **Edit the Database.sqlite by using for example "DB Browser for SQLite".**
> **Open the table "accounts" and set value to 0, 1, 2 or 3 in the "banned" column for an account.**

*ADDITIONAL STEPS, ONLY FOR THE SPECIAL VERSION

Step 27

Configure the MToolsSettings.asset in "Assets/uMMORPG/Scripts/Addons/ManagementTools/Resources"

Click the Load Items button (Templates/Item Manager). All items will be loaded automatically and divided by category. Your project will also be saved.

Make sure the Environments Folder path ends with "Special" (Templates/Creator). Click the Load Objects button. All monsters, npcs and environments will be loaded automatically and divided by category, then assigned to NetworkManager Registered Spawnable Prefabs. The Environment

Tag and Layer will be set automatically on all prefabs that are in the destination folder. Your project will also be saved.

You can always click the `Restore Defaults` button (at the bottom in the Inspector window) to restore default settings. Your project will also be saved.

> **MToolsSettings.asset stores all settings related to this addon. You can configure it however you want.**
>
> **You will also find an additional option here:**
> **Item Drop turn on if you have Item Drop. Now an additional field with items in the Database Information is active.**

Step 28

Go to World/Environment

Click the `Clear` button to clear any NavMesh.

Turn off NavMeshSurface in the Inspector window.

Remove any static properties of every prefabs (turn off Static field in the Inspector window) and add component NavMeshSourceTag.cs in the Inspector window of everything that need to be updated for the NavMesh, so objects related with a terrain (including it itself) or edit in the scene World/Environment as the following instruction:

Click on World/Environment and turn off Static field in the Inspector window on the top right corner, select “Yes, change children”.

Click on World/Environment/Structures/Steps to open its child objects, hold down the `Shift` key, click the last of them and add component NavMeshSourceTag.cs.

Act similarly with:

World/Environment/Structures/Planks

World/Environment/Structures/Fences

World/Environment/Structures/Logs

World/Environment/Structures/Walls

World/Environment/Rocks

World/Environment/Props/Boulders

World/Environment/Props/Wheels

World/Environment/Props/Buckets

World/Environment/Terrain

World/Environment/Trees

World/Environment/Buildings

Step 29

Go to World/PortalWithinRocks

Turn off Static field in the Inspector window on the top right corner, select “Yes, change children”.

Add component NavMeshSourceTag.cs in the Inspector window of every World/PortalWithinRocks‘s child objects except PortalToInstance.prefab.

Step 30

Drag and drop NavmeshArea.prefab from “Assets/uMMORPG/Scripts/Addons/ManagementTools/Prefabs/Special” to the World scene.

You can change the Size in the Inspector window of this prefab to match your terrain (click on your NavmeshArea.prefab on the scene to see the green lines).

Step 31

Create new account “admin”, will be with the owner access. Now every character on this account has the owner access.

Use the M key to open the Management Tools. Now you can change all logged players access in the panel to 0, 1 or 2. Go to the **table of contents** to find out information about all other features.

> **How to set the access level outside the game:**
> **Edit the Database.sqlite by using for example “DB Browser for SQLite”.**
> **Open the table “accounts” and set value to 0, 1, 2 or 3 in the “banned” column for an account.**